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| EX.NO:1 | Basic Networking Commands in Linux and Windows operating Systems |
|---|---|
| 17/07/2026 | |

**Aim:**

To study and execute various basic networking commands in the Windows and Linux operating systems to understand their purpose and output.

**Basic Networking commands in windows**

| command | Full Form / Purpose |
|---|---|
| arp -a | Address Resolution Protocol |
| hostname | Host Name |
| ipconfig /all | Internet Protocol configuration |
| nbtstat -a <hostname> | NetBIOS over TCP/IP Statistics |
| netstat -r | Network Statistics |
| nslookup www.google.com | Name Server Lookup |
| pathping www.google.com | Path + Ping |
| ping www.google.com | Packet Internet Grouper |
| route print | Route command |

## Basic Networking commands in Linux:

| command | Full Form / Purpose |
|---|---|
| ifconfig | Interface configuration |
| hostname | Host Name |
| ping www.google.com | Packet Internet Groper |
| traceroute www.google.com | Trace Route |
| netstat -r | Network statistics |
| nslookup www.google.com | Name Server LOOKUP |
| route -n | Route command |
| arp -n | Address Resolution Protocol |
| nmcli device show | Network Manager command Line Interface |
| ip addr show | IP Address command |

**Result:**

The basic networking commands were successfully executed in windows and Linux terminals. The systems IP configuration, DNS details, routing table and connectivity status were obtained correctly.

EXP.NO:2

# Learning and Assignment of IP Address manually to computers

**Aim:**

To configure and assign an IP address manually to a Wi-Fi adapter in the windows operating system.

**Procedure:**

1. Open Control Panel → Network and Internet → Network and Sharing center.
2. Click change Adapter settings on the left side.
3. Right-click on your Wi-Fi adapter → Properties.
4. Select Internet Protocol Version 4 (TCP/IPv4) and click Properties.
5. Select Use the following IP address and enter:
   - IP Address: 192.168.1.20
   - Subnet Mask: 255.255.255.0
   - Default Gateway: 192.168.1.1
   - Preferred DNS Server: 8.8.8.8
6. Click OK and close all windows.
7. Open Command Prompt and verify configuration:

None
ipconfig
ping 192.168.1.1
ping www.google.com

**Description:**

Manual IP assignment for Wi-Fi allows a user to configure a static IP address instead of using DHCP. This ensures consistent network addressing, better device management and improved troubleshooting in LAN/WLAN networks.

**Result:**

The Wi-Fi adapter was successfully configured with a manual IP address. The computer was able to communicate with the router and access the internet successfully.

| EX.NO: 3 | Study of different types of network cables and crimping of cable with RJ45 connector |
|---|---|

**Aim:**

To study different types of network cables and to prepare (crimp) a network cable using RJ45 connectors for establishing LAN connections between network devices.

**Apparatus Required:**

| S.NO | Equipment / component | Quantity |
|---|---|---|
| 1. | CAT 5e / CAT 6 UTP cable | 1 roll |
| 2. | RJ45 connectors (8P8C) | 2 nos |
| 3. | crimping tool | 1 |
| 4. | Cable stripper / cutter | 1 |
| 5. | LAN tester (optional) | 1 |
| 6. | Computer systems / Switch | As required |

**Theory:**

A network cable is a medium through which data is transmitted between computers and other network devices. Different types of network cables are used based on speed, distance and network environment.

1. **Types of Network Cables**

(i) **Twisted Pair cable**

- Most commonly used for LAN (Local Area Network).
- Consists of pairs of insulated copper wires twisted together.
- **Types:**
  - * UTP (Unshielded Twisted Pair) – No metallic shielding. Used in LAN.
  - * STP (Shielded Twisted Pair) – Shielded to reduce interference.

- Categories (CAT):

| Category | Speed | Frequency |
|---|---|---|
| CAT 5 | Up to 100 Mbps | 100 MHz |
| CAT 5e | Up to 1 Gbps | 100 MHz |
| CAT 6 | Up to 10 Gbps | 250 MHz |
| CAT 7 | Up to 10 Gbps | 600 MHz |

(ii) Coaxial Cable

- Has a central copper conductor, dielectric insulator and braided shield.
- Used in cable TV and early Ethernet.
- Types:
  - * ThinNet (10Base2)
  - * ThickNet (10Base5)

(iii) Fiber Optic Cable

- Uses light signals instead of electrical signals.
- Offers very high bandwidth and long-distance communication
- Types:
  - * Single Mode Fiber (SMF) - long distance, thin core (8-10 μm).
  - * Multi Mode Fiber (MMF) - short distance, thick core (50-62.5 μm).

2. Types of Ethernet Cables (Based on connection type)

| Type | Use | Description |
|---|---|---|
| Straight-Through Cable | connects PC → Switch / Router | Both ends follow same color code (T568B - T568B). |
| Crossover Cable | connects PC ↔ PC or Switch ↔ Switch | One end T568A, other end T568B |

3. RJ45 Connector:

* RJ = Registered Jack
* RJ45 is an 8-pin modular connector (8P8C) used for Ethernet cables.
* Each pin carries a specific signal in the network.

4. Color coding standards

(A) T568A color code

| Pin | Wire color | Function |
|---|---|---|
| 1. | White/Green | Transmit + |
| 2. | Green | Transmit - |
| 3. | White/Orange | Receive + |
| 4. | Blue | Unused |
| 5. | White/Blue | Unused |
| 6. | Orange | Receive - |
| 7. | White/Brown | Unused |
| 8. | Brown | Unused |

(B) T568B color code

| Pin | Wire color | Function |
|---|---|---|
| 1. | White/Orange | Transmit + |
| 2. | Orange | Transmit - |
| 3. | White/Green | Receive + |
| 4. | Blue | Unused |
| 5. | White/Blue | Unused |
| 6. | Green | Receive - |
| 7. | White/Brown | Unused |
| 8. | Brown | Unused |

5. Crimping Process

(A) For Straight-Through Cable

- Both ends follow the same standard (T568B - T568B).

(B) For Crossover Cable

- One end uses T568A, the other end T568B.

Step-by-Step Procedure:

1. Cut the UTP cable:

* Take required length (1 - 2 meters) of CAT5e/CAT6 UTP cable.

2. Strip the cable:

* Use wire stripper to remove about 1 inch of insulation from both ends.

3. Untwist and Arrange Pairs:

* Separate the 4 twisted pairs and arrange them according to the color code standard (T568A or T568B).

4. Trim the Ends:

* Cut all 8 wires evenly to the same length (about 1/2 inch from jacket).

Observation Table:

| Cable Type | End 1 code | End 2 code | Purpose | Result |
|---|---|---|---|---|
| Straight | T568B | T568B | PC-Switch | Working |
| Crossover | T568A | T568B | PC-PC | Working |

5. Insert wires into RJ45 connector:

* carefully insert the wires into RJ45 ensuring correct order and full contact with pins.

6. Crimp the connector:

* Place RJ45 into crimping tool and firmly press until the connector is properly crimped.

7. Repeat for other End:

* For straight cable: Use the same color code on both ends.

* For crossover cable: Use T568A on one end and T568B on the other.

8. Test the cable:

* Use a LAN tester or connect devices to verify connectivity (check link lights).

Result:

Different types of network cables were studied successfully. A UTP CAT5e cable was crimped using RJ45 connectors to form straight-through and crossover cables. Connectivity between network devices was successfully established and tested.

EX.NO:4 Implement packet sniffing using raw sockets in Python

Aim:

To capture and analyze various network protocol packets (like TCP, UDP, ARP, DNS, HTTP, ICMP, DHCP) using Wireshark and apply display filters to inspect them.

Software Used:

- Wireshark Network Protocol Analyzer

Procedure:

1. Open Wireshark
2. Select Local Area connection (or your active interface).
3. Go to Capture → Options.
4. Set: Stop capture automatically after 100 packets.
5. Click Start to begin capturing.
6. Use filter bar to type specific protocols like tcp, arp, etc.
7. Use Statistics → Flow Graph (for TCP/DNS) to view packet flow.
8. Click File → Save to store the captured packets.

Tasks & Outputs

1. Filter: Display TCP/UDP packets and Flow Graph.
   * Filter used: tcp or udp
   * Output: Displayed only TCP/UDP packets.
   * Flow Graph: viewable from Statistics → Flow Graph.
2. Filter: Display only ARP packets
   * Filter used: arp
   * Output: Displayed only Address Resolution Protocol packets.

3. Filter: Display only DNS packets and Flow Graph

* Filter used: dns
* Output: Displayed only DNS queries and responses.
* Flow Graph: Shows DNS request - response communication

4. Filter: Display only HTTP packets

* Filter used: http
* Output: Captured HTTP traffic (browser or server communication).

5. Filter: Display IP/ICMP packets

* Filter used: icmp or ip
* Output: Shows ICMP packets (used in ping operations) and IP packet info.

6. Filter: Display only DHCP packets

* Filter used: bootp (used for DHCP)
* Output: Displays DHCP Discover, Offer, Request, Acknowledgement packets.

Result:

Successfully captured and filtered different protocol packets using Wireshark and understood how to inspect and analyze them with tools like filters and flow graphs.

| EX.NO: 5 | Develop a customized ping command to test the server connectivity |
|---|---|

**Aim:**

To test and verify the connectivity between the local computer and remote servers using the ping command in windows

**Description:**

The ping command is a basic network utility used to check whether a host is reachable across an IP network. It measures the round-trip time for messages sent from the source to a destination and reports packet loss and latency.

**Procedure:**

1. Basic Ping Command:

ping www.google.com

O/P: Displays the IP address of the server, the number of packets sent, received, lost and the average round-trip time.

2. Ping Localhost (to check Internal network stack) Command:

ping localhost

O/P: Replies from 127.0.0.1 confirm that the TCP/IP stack on the local system is working properly.

3. Ping Using IP Address command:

ping 8.8.8.8

O/P: Displays connectivity and delay time (in milliseconds) between your system and Google DNS server.

4. Continuous Ping Test Commands:

ping www.google.com -t

O/P: Continuously pings the server until stopped manually (ctrl+c), showing live latency and packet information.

5. Ping with specific Number of Echo Requests command:

ping www.google.com -n 5

O/P: Sends exactly 5 echo requests and displays the result with average time and packet loss summary.

6. Ping with Packet Size Specification

ping www.google.com -l 1000

O/P: Sends a ping request with a packet size of 1000 bytes, showing transmission performance.

Result:

The ping commands were successfully executed in the windows command Prompt. The network connectivity, latency and packet transmission status between the local computer and various servers were verified.

EX.NO:9 | Design a simple topology and configure with one router, two switches and PCs using cisco Packet Tracer

Aim:

To design and configure a simple network topology consisting of one router, two switches and multiple PCs in Cisco Packet Tracer and enable communication between PCs.

Equipment needed:

* Cisco Packet Tracer Software
* 1 Router
* 2 switches
* 4 PCs
* Appropriate cables (copper straight-through)

Procedure:

1. open cisco Packet Tracer
2. Add 1 Router (2911) to the workspace
3. Add 2 switches (2960) to the workspace
4. Add 4 PCs to the workspace.
5. Select copper Straight-Through cable and make the following connections:
   - PC1 → Switch 1 Fa0/1
   - PC2 → Switch 1 Fa0/2
   - Switch 1 Fa0/24 → Router G0/0
   - Router G0/1 → Switch 2 Fa0/24
   - Switch 2 Fa0/1 → PC3
   - Switch 2 Fa0/2 → PC4
6. Open Router → CLI and enter:

```
enable
configure terminal
interface gigabitEthernet 0/0
ip address 192.168.1.1 255.255.255.0
no shutdown
```

exit
interface gigabitEthernet 0/1
ip address 192.168.2.1 255.255.255.0
no shutdown
exit
End

7. Configure PC1 from Desktop → IP configuration:
   - IP: 192.168.1.2
   - Subnet Mask: 255.255.255.0
   - Gateway: 192.168.1.1

8. Configure PC2:
   - IP: 192.168.1.3
   - Subnet Mask: 255.255.255.0
   - Gateway: 192.168.1.1

9. Configure PC3:
   - IP: 192.168.2.2
   - Subnet Mask: 255.255.255.0
   - Gateway: 192.168.2.1

10. Configure PC4:
   - IP: 192.168.2.3
   - Subnet Mask: 255.255.255.0
   - Gateway: 192.168.2.1

11. Open PC1 → Desktop → Command Prompt

12. Test the connection by entering:

    ping 192.168.2.2

13. If required, save the router configuration:

enable
copy running-config startup-config

Result:

The PCs connected to different switches can communicate with each other via the router. The successful ping replies verify that the network is properly configured and operational

Ex. No 10 Customize switch with network modules using Cisco Packet Tracer

Aim:

To learn how to add a network module to a Cisco switch, connect multiple PCs and a server, assign IP addresses and test connectivity in Cisco Packet Tracer.

Description:

This experiment is about building a simple computer network using a switch with a module ports. You will connect five PCs and one server, assign IPs and check if all devices can communicate. This helps you understand how modular switches are used to create flexible networks.

Procedure:

1. Open Cisco Packet Tracer and set up your workspace.
2. Drag and drop a 2960-24TT switch to the center.
3. Add five PCs and one Server-PT from the End Devices panel.
4. Power off the switch:
- Click the switch, open the physical tab and turn off the power button.
5. Install HWIC-4ESW module on the switch:
- In the Physical tab, drag HWIC-4ESW into an empty module slot.
6. Power on the switch and let it boot.
7. Connect all devices using Copper Straight-through cables:
- PC0 → Switch FastEthernet 0/1
- PC1 → Switch FastEthernet 0/2
- PC2 → Switch Module Port 1
- PC3 → Switch Module Port 2
- PC4 → Switch Module Port 3 - Server0 → Switch Module Port 4
8. Assign IP addresses to each device:

Result: Thus, the program was executed successfully